# Scientific data pipeline

Discover → snapshot → validate → normalize → derive → review → publish.

PROPOSED • Not implemented • Scientific goal, scoring weights and hosting provider remain undecided

01

## Source registry and ingestion

Select specific products from multiple NASA missions for Jezero: candidate sources include MRO imagery, MGS topography and Perseverance observations. Verify coverage and alignment. Record product IDs, citations, timestamps, licenses and checksums.

02

## Planetary coordinate handling

Record body, CRS, latitude type, longitude direction/range, height datum and units for each product. Normalize explicitly; handle the longitude seam and poles. Do not label Mars positions EPSG:4326 or use Earth distances by default.

03

## Raster processing

Retain original DEM and imagery. Produce validated analysis rasters and separate display tiles. GDAL COG is a proposed storage format; COG is not a ready-to-use Cesium terrain tileset. Derive slope only with verified horizontal and vertical units.

04

## Core data entities

craters/regions: discovery context. stops: verified coordinates, science tasks, tools and work time. datasets/observations: provenance, resolution and values. excursions: base, ordered stops and return. runs: input, model and dataset versions.

05

## Coverage and quality gates

Check duplicates, units, coordinate transforms, nodata, raster alignment and valid geometry. Treat footprints and pixel resolution as eligibility inputs. Sample-level coordinates remain pending until verified; a rover location is not automatically a drill location.

06

## Serving and refresh

Publish reviewed dataset versions atomically. Store large rasters outside database rows; index footprints and metadata in PostGIS. Serve prebuilt display tiles via storage/CDN. Refresh deliberately, retaining old versions for reproducible rankings and rollback.

Read technical-brief.md for linked sources, assumptions and competition timing.